Sam Gowel

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Education

Virginia Tech Blacksburg, Virginia
Aerospace Engineering — GPA: 3.68 — Dean's List: 25/26 — Junior Level Course Work *Fall 2025 – Spring 2029*

Relevant Experience

R&D Compisites Lead February 2026 – Present
SolarCar at VT

- Led the redesign of the aeroshell's internal rib structure and top-shell hinging mechanism, projected to reduce component cost by $\sim 80\%$ and access time from ~ 2 minutes with 8 people to ~ 30 seconds with 1 person.
- Analyzed custom designs for the hinging mechanism through SolidWorks CAD, Finite Element Analysis, as well as MATLAB to evaluate feasibility and ensure manufacturability.
- Prototyped and manufactured components using 3D printing, laser cutting, carbon fiber layups, and hand tools, integrating digital designs with hands on fabrication.
- Working with carbon fiber FEA with an aim to manufacture fully carbon fiber rims and a carbon fiber monocoque chassis.

Undergraduate Researcher, Recruitment Lead October 2025 – Present
VT E2E, Sustainable Aviation Fuel Research Group

- Developed a sustainable aviation fuel research topic by conducting literature reviews and designing experimental trials and test rigs to validate our approach.
- Helped the team earn the organization's award for most progress among 5 research groups through research, experimental design, and technical development.
- Investigated the effects of sustainable aviation fuel on dielectric constant and fuel sensor accuracy.
- Presented our work at 3+ conferences as well as bi-weekly technical reports to 30+ peers and professional mentors.
- Led recruitment efforts for the research group by coordinating outreach and finding new members.

Research Assistant March 2026 – May 2026
Virginia Tech College of Business

- Interacted with 40+ research participants across multiple groups while following experimental procedures and maintaining consistent data collection practices.
- Collaborated with research directors during debriefing meetings to guide trials and plan future research.

Projects

Inverted Pendulum Portfolio: samgowel.com

- Designed and built an inverted pendulum that is able to balance itself using a custom PD controller.

12-DOF Quadruped Robot Portfolio: samgowel.com

- Designed and built a 12-DOF quadruped robot using custom CAD, 3D-printed components, servo motors, and inverse kinematics. Developed a crawl gait and servo control system to coordinate movement across legs.

Additional Experience

Customer Associate May 2026 – August 2026
Commissary

- Assisted up to 50 customers per hour while working collaboratively with team members to provide efficient and friendly service.
- Maintained a clean, organized, and safe checkout area while handling carts, returns, sanitation, and other daily tasks.

Skills

Programming: Python, Java, MATLAB
CAD/Simulation: SolidWorks, Fusion 360, Siemens NX, FEA, CFD, ANSYS
Manufacturing: 3D Printing, Carbon Fiber, Laser Cutting, Prototyping
Other: Microsoft Office Suite, LaTeX